

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Topic: Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

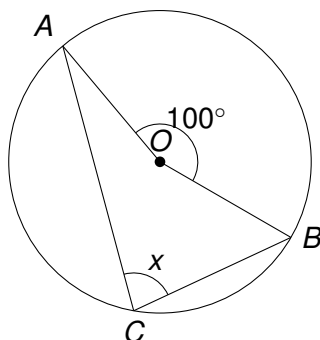
Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A space agency uses a circular satellite dish, part of a tracking array, to monitor deep space signals. Points A , B and C lie on the circumference of the dish's circular rim, with centre O .

The angle at the centre, AOB , is 100° .



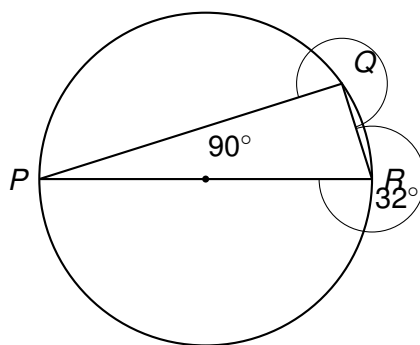
Work out the size of angle x , the angle at the circumference subtended by the same arc AB .

$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 1 is 2 marks)

- 2 A circular observation window on a space station is designed so that a triangular support brace PQR is fitted inside it, with PR as a diameter of the circle, as shown in the diagram.



PR is a diameter of the circle.

Angle $QRP = 32^\circ$.

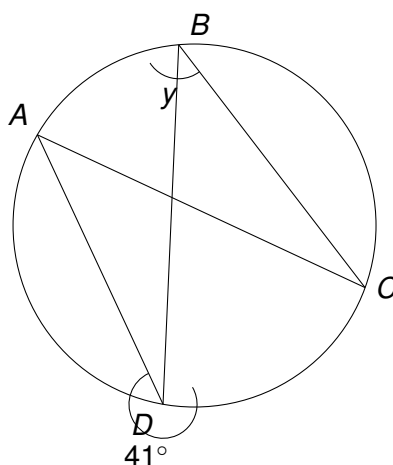
Find the size of angle QPR .

$QPR = \dots\dots\dots^\circ$

(2)

(Total for Question 2 is 2 marks)

- 3 An astronomer is studying a circular observation dome. Points A , B , C and D lie on the circumference of the circular base of the dome, as shown in the diagram below. Chords AC and BD are drawn, and lines AD and BC are also drawn.



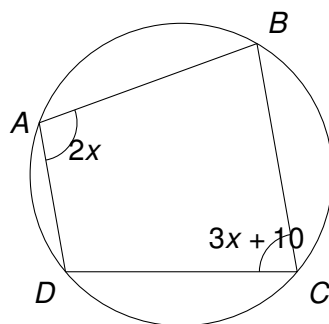
Angle $ADC = 41^\circ$.

Find the value of y , giving a reason for your answer.

$y = \dots\dots\dots$
(3)

(Total for Question 3 is 3 marks)

- 4 A , B , C and D are four points on the circumference of a circle, forming the cyclic quadrilateral $ABCD$. Angle $DAB = 2x$ and angle $BCD = 3x + 10$, as shown in the diagram.



- (a) Form an equation in terms of x , and use it to find the value of x .

$$x = \dots\dots\dots$$

(2)

- (b) Work out the size of angle DAB and angle BCD .

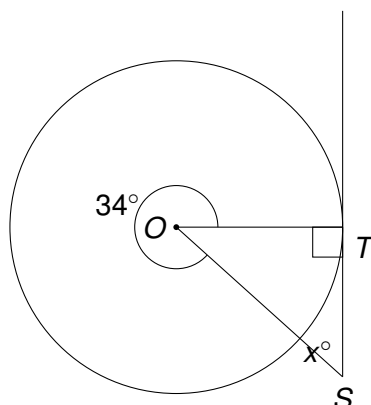
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(2)

(Total for Question 4 is 4 marks)

- 5 A satellite ground station uses a circular dish reflector to receive signals from a spacecraft during a deep-space mission. The signal path from the receiver arm meets the edge of the circular reflector at point T , touching it at exactly one point, so that the receiver arm ST is a tangent to the circle with centre O .

OT is a radius of the reflector, and OS is a support strut connecting the centre of the dish to a fixing point S on the receiver arm. The angle between the support strut OS and the radius OT is 34° .



Work out the size of angle OST , marked x° on the diagram, giving a reason for each stage of your working.

$$x = \dots\dots\dots^\circ$$

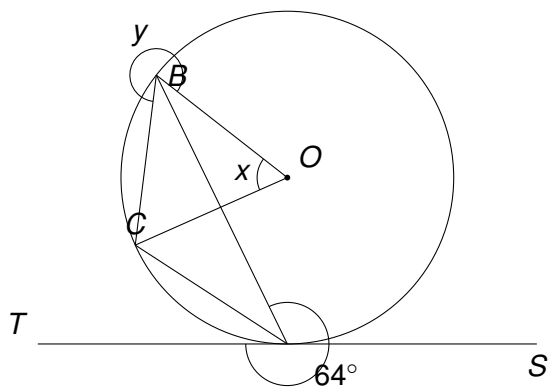
(3)

(Total for Question 5 is 3 marks)

6 A , B and C are points on a circle with centre O . The line TAS is a tangent to the circle at A . OB and OC are radii of the circle.

Angle $BAT = 64^\circ$.

O is the centre of the circle, so $OA = OB = OC$ (radii of the same circle).



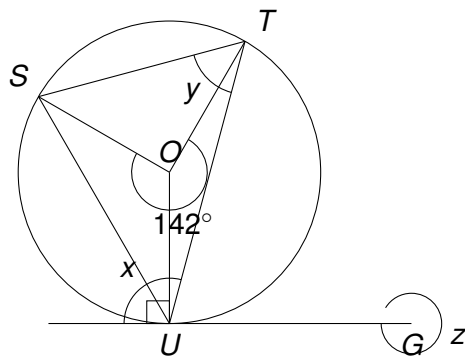
Show that angle OBC (marked y) is equal to angle BOC split in a way that gives $y = 26^\circ$, by showing that triangle OBC is isosceles with the base angles each equal to 26° .

(5)

(Total for Question 6 is 5 marks)

- 7 A space agency tracks a satellite moving around a circular orbit. Points S , T and U lie on the circular orbit path, with centre O (the centre of the Earth's projected orbit plane). A ground station at point G lies on the tangent to the orbit at U , so that GU is a tangent to the circle at U .

$STUO$... in the diagram below, $OSTU$ forms part of the configuration, where O is the centre of the circle, S , T , U lie on the circumference, and $STUG$ is completed by the tangent line at U meeting the extended chord at G .



O is the centre of the circular orbit. S , T and U are three satellite positions on the orbit. The angle at the centre, $\angle SOT = 142^\circ$. GU is a tangent to the circle at U , and G , U , S do not lie on a straight line through O . $STUG$ is arranged so that SU extended does not pass through G ; instead $\angle UGS = z$ is the angle between the tangent GU and the chord US extended to meet the tangent line at G (assume G , S and the relevant extended chord are positioned such that triangle GUS is formed, with GU tangent at U).

- (a) Find the size of angle STU (y), the angle subtended by arc SU at the circumference (not containing T , i.e. the angle in the major arc).

$$y = \dots\dots\dots^\circ$$

(2)

- (b) $OSTU$... O , S , T , U together with the tangent do not form a cyclic quadrilateral directly; instead, consider quadrilateral $OSTU$ is not cyclic since O is the centre. Using the fact that triangle OST is isosceles (as $OS = OT$, both radii), find the angle OST .

$$\angle OST = \dots\dots\dots^\circ$$

(2)

- (c) Given that OU is a radius and GU is a tangent at U , and that triangle OUS is isosceles with $OU = OS$, use the tangent-radius property together with your answers above to find the angle $z = \angle UGS$ between the tangent GU and the chord US .

$$Z = \dots\dots\dots^\circ$$

(2)

(Total for Question 7 is 6 marks)

TOTAL FOR PAPER IS ?? MARKS